CREATE++: Extending Cross-Representation Alignment with Signed and Higher-Order Signals

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***Abstract*—Transformer architectures have become the dominant approach in sequential recommender systems due to their ability to capture dependencies within user interaction histories. However, these models often consider sequence elements in isolation, failing to fully leverage the global collaborative context inherent in user–item relationships. While recent frameworks like CREATE [1] address this by aligning sequential encoders with graph neural networks (GNNs), they remain largely limited to unsigned bipartite graphs that cannot model explicit negative feedback or higher-order multi-way relationships. To fill this gap, we propose two architectural extensions: CREATE-Pone and CREATE-Uni. By integrating signed graph modeling inspired by Pone-GNN [2] to disentangle interest from disinterest and hypergraph structures via the UniGNN [3] framework to capture complex higher-order correlations, our framework provides a more expressive and nuanced grounding for sequential models. This dual approach significantly enriches the structural knowledge transferred to the transformer while preserving the efficient asymmetric inference protocol required for production-scale recommendation.**

***Index Terms*—Recommender Systems, Sequential Recommendation, Signed Graphs, Hypergraphs**

I. Introduction

Modern recommender systems must reconcile two distinct perspectives of user behavior: short-range sequential dynamics and long-range collaborative structures. While Graph Neural Networks (GNNs) have emerged as the state-of-the-art for capturing global collaborative signals through recursive neighborhood aggregation, they are often less effective at modeling the strictly ordered, temporal dependencies inherent in user interaction histories. Conversely, Transformer-based architectures excel at capturing these fine-grained sequential patterns but struggle to encode the broader structural context of the global user–item interaction graph. To bridge this gap, the recently proposed CREATE framework [1] introduces a paradigm of cross-representation knowledge transfer. By using a GNN as a global structural grounding to enrich a transformer-based sequential encoder during training, CREATE achieves superior performance by combining the strengths of both architectures. This alignment allows the model to inherit global collaborative wisdom while maintaining an efficient asymmetric inference protocol that utilizes only the sequential encoder at test time.

Despite its effectiveness, the original CREATE framework relies on unsigned bipartite graphs, which introduce two significant structural bottlenecks. First, it suffers from *polarity blindness*; by treating all observed interactions as positive, it ignores explicit negative feedback (e.g., low ratings or skips), leading to label noise and an inability to model user aversion. Second, it is limited by *pairwise collapse*, where higher-order multi-way relationships—such as a user's entire session history or co-purchase groups—are reduced to isolated binary links. This simplification results in a loss of complex collaborative context that pairwise edges cannot distinguish. Third, the framework is hindered by *exposure bias*. Because unsigned bipartite GNNs typically treat all unobserved interactions as negative signals, the model fails to distinguish between items a user actively disliked and those they simply never encountered, leading to suboptimal generalization.

In this paper, we propose two extensions to the CREATE framework to recover this lost structural information:

- **CREATE-Pone (Signed Variant):** We replace the unsigned encoder with a signed GNN [2] that learns dual interest and disinterest embeddings. The alignment framework is extended to a multi-objective task: the sequential user representation is aligned with the global interest view to maximize knowledge transfer, while simultaneously being constrained to be orthogonal to the disinterest view. This allows the transformer to inherit collaborative preferences while explicitly suppressing signals associated with user aversion.
- **CREATE-Uni (Hypergraph Variant):** We leverage a unified hypergraph framework [3] to model user histories as hyperedges. This transition enables the model to capture multi-way collaborative signals with 1-GWL expressive power, strictly exceeding the topological limits of traditional bipartite GNNs like LightGCN.

Our contributions include the formalization of these signed and higher-order alignment objectives, a theoretical analysis of their expressive advantages, and an extensive empirical evaluation across five benchmark datasets. The remainder of this paper is organized as follows: Section II reviews related work, Section III presents the methodology, Section IV provides theoretical analysis, and Section V describes the experimental setup.

II. Related Work

The evolution of recommendation systems has moved from modeling isolated sequences to capturing global structural dependencies, and recently, toward the alignment of these two distinct paradigms.

A. Sequential Recommendation

Sequential recommendation focuses on predicting the next item based on a user's chronological interaction history. Early models utilized Recurrent Neural Networks (RNNs), but the field has been revolutionized by Transformer-based architectures. *SASRec* [4] adopted a self-attention mechanism to identify relevant items in a sequence, while *BERT4Rec* [5] introduced bidirectional modeling via masked language objectives. To improve robustness, *CL4SRec* [6] integrated contrastive learning to ensure that different augmentations of the same sequence yield similar representations. However, these models primarily focus on local dependencies and often lack a global perspective of item-item collaborative signals.

B. Graph-based Collaborative Filtering

Graph Neural Networks (GNNs) explicitly model user–item interactions as a bipartite graph to capture long-range collaborative signals. This paradigm involves learning d-dimensional node embeddings for both users ($\mathbf{e}_u$) and items ($\mathbf{e}_i$) by aggregating information from their respective neighborhoods. *NGCF* [7] first introduced neighborhood aggregation to recommendation, but was later simplified by *LightGCN* [8], which removed non-linear activations and weight matrices to improve efficiency and performance:

$$\mathbf{e}_u^{(k+1)} = \sum_{i \in \mathcal{N}(u)} \frac{1}{\sqrt{|\mathcal{N}(u)||\mathcal{N}(i)|}} \mathbf{e}_i^{(k)} \tag{1a}$$

$$\mathbf{e}_i^{(k+1)} = \sum_{u \in \mathcal{N}(i)} \frac{1}{\sqrt{|\mathcal{N}(i)||\mathcal{N}(u)|}} \mathbf{e}_u^{(k)} \tag{1b}$$

After K layers of propagation, the final embeddings are typically a weighted sum or average of the embeddings at each layer. Predicted scores are then computed via the inner product between the final user and item embeddings: $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$.While GNNs are highly effective for capturing global structural patterns, they suffer from two primary issues in dynamic settings. First, they are inherently limited by the 1-dimensional Weisfeiler-Leman (1-WL) test, meaning they primarily capture pairwise dependencies and fail to recognize higher-order multi-way correlations. Second, they face the folding-in problem: generating recommendations for a new user or a new session at inference time requires re-running expensive message-passing operations across the global graph or retraining the model entirely. GNNs typically lack the flexibility to handle the real-time evolution of user interests due to this high computational cost at test time. To avoid this bottleneck, modern hybrid systems utilize sequential models for inference, relying on the GNN only during the training phase to provide an enriched, structural grounding for the sequential item embeddings.

C. Cross-Representation Knowledge Transfer

The *CREATE* framework [1] serves as the primary basis for this work, bridging the gap between local sequential patterns and global collaborative structures through representation alignment. The framework utilizes a dual-encoder architecture: a sequential encoder (such as SASRec [4] or BERT4Rec [5]) processes the interaction history to generate a temporal user representation $\mathbf{h}_u$, while a graph encoder (typically LightGCN [8] or UltraGCN) processes the global bipartite graph to produce structural user embeddings $\mathbf{z}_u$ and item embeddings $\mathbf{z}_i$.The core innovation of CREATE is the use of the *Barlow Twins* objective [9] to align these two distinct views. After batch standardization of $\mathbf{h}_u$ and $\mathbf{z}_u$, the model computes a cross-correlation matrix $\mathcal{C}$, where the alignment loss $\mathcal{L}_{align}$ forces the sequential representation to inherit structural context by driving $\mathcal{C}$ toward the identity matrix:

$$\mathcal{L}align = \sum_i (1 - \mathcal{C}ii)^2 + \lambda \sum_{i \neq j} \mathcal{C}ij^2 \tag{2}$$

The first term ensures invariance (alignment), while the second term minimizes redundancy between the dimensions. To ensure the sequential encoder effectively learns from the structural grounding, the framework adopts a joint training objective:

$$\mathcal{L} = \mathcal{L}local + \omega_{global}\mathcal{L}global + \omega align\mathcal{L}align \tag{3}$$

Here, $\mathcal{L}local$ represents the next-item prediction loss for the sequential model, $\mathcal{L}_{global}$ is the recommendation loss for the GNN (e.g., BPR), and ω are weighting hyperparameters. Predicted scores are then computed via the inner product between the sequential user and graph item embeddings: $\hat{y}_{ui} = \mathbf{h}_u^\top \mathbf{z}_i$, thus the model eliminates the need for the graph encoder during inference, allowing for efficient real-time recommendation using only the sequential encoder. Our work extends this objective by replacing the standard bipartite GNN with more expressive signed and hypergraph encoders.

D. Hypergraph Modeling and UniGNN

To capture multi-way relationships (e.g., items co-purchased in a single session), hypergraph-based models have emerged. *HGNN* [10] and *HNHN* [11] provide frameworks for high-order correlation modeling. *UniGNN* [3] generalized these approaches into a two-stage message-passing framework:

$$\begin{cases} \mathbf{h}_e = \phi_1\left(\{\mathbf{x}_j\}_{j \in e}\right) & \text{(Node to Hyperedge)} \\ \tilde{\mathbf{x}}_i = \phi_2\left(\mathbf{x}_i, \{\mathbf{h}_e\}_{e \in E_i}\right) & \text{(Hyperedge to Node)} \end{cases} \tag{4}$$

Furthermore, *DHCN* [12] demonstrated that modeling sessions as hyperedges allows the capture of higher-order signals that traditional graphs collapse into pairwise links. Our *CREATE-Uni* extension leverages this 1-GWL expressive power to enrich sequential grounding.

E. Signed Graph Learning

Despite the success of alignment frameworks, most GNN-based recommenders operate on unsigned bipartite graphs, treating all observed interactions as positive signals. This introduces significant label noise and exposure bias, as the model cannot distinguish between items a user actively disliked and those they simply never encountered. To address this, *Pone-GNN* [2] introduces a signed graph framework that explicitly models negative feedback by partitioning the

global interaction set $\mathcal{E}$ into two disjoint subsets: a positive interaction graph $\mathcal{G}_p = (\mathcal{V}, \mathcal{E}_p)$ and a negative interaction graph $\mathcal{G}_n = (\mathcal{V}, \mathcal{E}_n)$.The framework employs two independent message-passing branches to learn dual representations for each node: an interest embedding $\mathbf{z} \in \mathbb{R}^d$ derived from $\mathcal{G}_p$, and a disinterest embedding $\mathbf{v} \in \mathbb{R}^d$ derived from $\mathcal{G}_n$. These node embeddings are optimized via a contrastive learning loss that ensures the latent spaces are well-separated and informative. For the final recommendation task, the model utilizes a dual feedback-aware loss $\mathcal{L}_{DF}$ to maximize the likelihood of observed interactions within their respective polarities:

$$\mathcal{L}_{\mathrm{DF}} = -\sum_{(u,i)\in\mathcal{E}_p} \log\sigma\big(\mathbf{z}_u^\top \mathbf{z}_i\big) - \sum_{(u,j)\in\mathcal{E}_n} \log\sigma\big(-\mathbf{v}_u^\top \mathbf{v}_j\big)\,. \quad (5)$$

Our *CREATE-Pone* extension integrates this signed modeling into the alignment framework, enabling the sequential encoder to explicitly suppress aversion signals.

III. Proposed Methodology

Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items, respectively. Each user $u \in \mathcal{U}$ has a timestamped interaction history $H_u = (i_1^u, \ldots, i_{|H_u|}^u)$. We retain the shared embedding layer, the sequential encoder (SASRec with full softmax cross-entropy loss L_{local}), and the two-phase training procedure (graph/hypergraph warm-up followed by joint optimization) from the original CREATE framework.

A. CREATE-Pone: Signed Graph Variant

We partition $\mathcal{E}$ into positive interactions $\mathcal{E}_p$ (ratings ≥ 4 or explicit likes) and negative interactions $\mathcal{E}_n$ (ratings ≤ 3 or explicit dislikes), yielding two disjoint graphs. This fixed threshold is a practical initialization; recent work such as ReFINe and ReFINe++ [13], [14] suggests LLM-assisted refinement can reclassify ambiguous low ratings by modeling user-specific preference patterns. Exploring these adaptive heuristics is a promising future direction:

$$\mathcal{G}_p = (\mathcal{U}, \mathcal{I}, \mathcal{E}_p), \quad \mathcal{G}_n = (\mathcal{U}, \mathcal{I}, \mathcal{E}_n).$$

Each node $v \in \mathcal{U} \cup \mathcal{I}$ is assigned two initial embeddings:

$$\mathbf{z}_v^{(0)}, \mathbf{v}_v^{(0)} \in \mathbb{R}^d$$

(interest from $\mathcal{G}_p$, disinterest from $\mathcal{G}_n$).

We construct two training sets for mini-batch learning:

$$D_p = \{(u,i,j) \mid (u,i) \in \mathcal{E}_p,\ j \notin \mathcal{N}_p(u),\ u \in \mathcal{U},\ i,j \in \mathcal{I}\}, \quad (6a)$$

$$D_n = \{(u,i,j) \mid (u,i) \in \mathcal{E}_n,\ j \notin \mathcal{N}_n(u),\ u \in \mathcal{U},\ i,j \in \mathcal{I}\}. \quad (6b)$$

Mini-batches sampled from these sets are denoted by $B_p \subset D_p$ and $B_n \subset D_n$, respectively.

For layers $k = 0, \ldots, K-1$, we perform dual-branch propagation on both signed graphs.

On the positive graph $\mathcal{G}_p$ and negative graph $\mathcal{G}_n$, updates are

$$\mathbf{z}_u^{(k+1)} = \frac{1+\epsilon_p^{(k)}}{|\mathcal{N}_p(u)|}\mathbf{z}_u^{(k)} + \sum_{i\in\mathcal{N}_p(u)} \frac{\mathbf{z}_i^{(k)}}{\sqrt{|\mathcal{N}_p(u)|\,|\mathcal{N}_p(i)|}}, \quad (7a)$$

$$\mathbf{z}_i^{(k+1)} = \frac{1+\epsilon_p^{(k)}}{|\mathcal{N}_p(i)|}\mathbf{z}_i^{(k)} + \sum_{u\in\mathcal{N}_p(i)} \frac{\mathbf{z}_u^{(k)}}{\sqrt{|\mathcal{N}_p(i)|\,|\mathcal{N}_p(u)|}}, \quad (7b)$$

$$\mathbf{v}_u^{(k+1)} = \frac{1+\epsilon_n^{(k)}}{|\mathcal{N}_n(u)|}\mathbf{v}_u^{(k)} + \sum_{i\in\mathcal{N}_n(u)} \frac{\mathbf{v}_i^{(k)}}{\sqrt{|\mathcal{N}_n(u)|\,|\mathcal{N}_n(i)|}}, \quad (7c)$$

$$\mathbf{v}_i^{(k+1)} = \frac{1+\epsilon_n^{(k)}}{|\mathcal{N}_n(i)|}\mathbf{v}_i^{(k)} + \sum_{u\in\mathcal{N}_n(i)} \frac{\mathbf{v}_u^{(k)}}{\sqrt{|\mathcal{N}_n(i)|\,|\mathcal{N}_n(u)|}}. \quad (7d)$$

In Eq. (7)a and (7)b are computed on the positive graph $\mathcal{G}_p$, while (7)c and (7)d are computed analogously on the negative graph $\mathcal{G}_n$. The final embeddings are layer-averaged:

$$\mathbf{z}_v = \frac{1}{K}\sum_{k=0}^{K-1} \mathbf{z}_v^{(k)}, \quad \mathbf{v}_v = \frac{1}{K}\sum_{k=0}^{K-1} \mathbf{v}_v^{(k)}. \quad (8)$$

The global loss is defined as follows.

- Dual feedback-aware loss L_{DF}:

$$L_{DF} = -\sum_{(u,i,j)\in B_p} \ln\sigma(\hat{y}_{u,i} - \hat{y}_{u,j}) - \sum_{(u,i,j)\in B_n} \ln\sigma(\hat{y}_{u,j} - \hat{y}_{u,i})\,, \quad (9)$$

  with

$$\hat{y}_{u,i} = \begin{cases} \mathbf{z}_u^\top \mathbf{z}_i, & \text{if } (u,i,j) \in B_p, \\ b \cdot \mathbf{v}_u^\top \mathbf{v}_i, & \text{if } (u,i,j) \in B_n, \end{cases}$$
$$\hat{y}_{u,j} = \begin{cases} \mathbf{z}_u^\top \mathbf{z}_j, & \text{if } (u,i,j) \in B_p, \\ \mathbf{v}_u^\top \mathbf{v}_j, & \text{if } (u,i,j) \in B_n. \end{cases} \quad (10)$$

  Here, $\hat{y}_{u,i}$ denotes the predicted similarity score between user u and item i.
- Contrastive learning loss L_{CL}:

$$L_{CL} = -\frac{1}{N} \sum_{\substack{(u,i,j)\in B_p \\ (u,i',j')\in B_n}} \log \frac{\exp(\mathbf{z}_u^\top \mathbf{z}_i/\tau)}{\exp(\mathbf{z}_u^\top \mathbf{z}_i/\tau) + \exp(\mathbf{v}_u^\top \mathbf{v}_{i'}/\tau)}. \quad (11)$$

The total global loss is

$$L_{\text{global}} = L_{DF} + L_{CL} \quad (12)$$

For user sequence $S_u = (i_1, \ldots, i_t)$:

$$\mathbf{x}_k = \mathbf{z}_{i_k} + \mathbf{p}_k, \quad \mathbf{h}_u = \text{Transformer}(\mathbf{x}_1, \ldots, \mathbf{x}_t)[t], \quad (13)$$

where only interest embeddings $\mathbf{z}_{i_k}$ are used as input, and $\mathbf{p}_k$ is the learnable positional embedding for the k-th position in the sequence. The transformer loss is the standard full softmax cross-entropy:

$$L_{\text{local}} = -\sum_{(u,t)} \log \frac{\exp(\mathbf{h}_u^\top \mathbf{z}_{i_{t+1}})}{\sum_{j\in\mathcal{I}} \exp(\mathbf{h}_u^\top \mathbf{z}_j)}. \quad (14)$$

Here, j indexes all candidate items in the item set $\mathcal{I}$.

After batch standardization of $\tilde{\mathbf{h}}_u$, $\tilde{\mathbf{z}}_u$, $\tilde{\mathbf{v}}_u$:

$$\mathcal{L}_{\text{align}} = \sum_i (1 - C^{hz}_{ii})^2 + \lambda \sum_{i \neq j} (C^{hz}_{ij})^2 + \mu \sum_{i,j} (C^{hv}_{ij})^2, \tag{15}$$

where C^{hz} and C^{hv} are cross-correlation matrices between batch-normalized representations $(\tilde{\mathbf{h}}_u, \tilde{\mathbf{z}}_u)$ and $(\tilde{\mathbf{h}}_u, \tilde{\mathbf{v}}_u)$, respectively; the first two terms align $\mathbf{h}_u$ with interest $\mathbf{z}_u$ (Barlow Twins style), and the third enforces orthogonality to disinterest $\mathbf{v}_u$.

The total loss is

$$\mathcal{L} = L_{\text{local}} + w_{\text{global}} L_{\text{global}} + w_{\text{align}} \mathcal{L}_{\text{align}}. \tag{16}$$

Training proceeds in two phases: (1) Warm-up — train only Pone-GNN using L_{global}; (2) Joint — end-to-end optimization of $\mathcal{L}$.

B. CREATE-Uni: Hypergraph Variant

Let U be the set of users, I the set of items and $V = U \cup I$. Each observed interaction is a pair (u, i, τ) where $u \in U$, $i \in I$ and $\tau \in \mathbb{R}$ is the interaction timestamp.

Fix a session length $t > 0$ (hours). For every user u we partition u's timeline into s_u contiguous sessions of length t. Denote the r-th session time interval for user u by

$$T_{u,r} = [\tau_0^{(u)} + (r-1)t, \;\; \tau_0^{(u)} + rt], \qquad r = 1, \ldots, s_u$$

where $\tau_0^{(u)}$ is a chosen reference time for u (e.g. first interaction or a global anchor). The set of items that u interacted with during session r is

$$H_{u,r} \coloneqq \{\, i \in I \mid \exists\, \tau \in T_{u,r} \text{ s.t. } (u, i, \tau) \text{ observed} \,\}.$$

Define the session hyperedge (for user u, session r) as

$$e_{u,r} \coloneqq \{\, u \,\} \cup H_{u,r}.$$

The hypergraph is then

$$\mathcal{H} = (V, \mathcal{E}), \qquad \mathcal{E} \coloneqq \big\{ e_{u,r} \mid u \in U,\; r = 1, \ldots, s_u,\; H_{u,r} \neq \varnothing \big\}.$$

We apply the two-stage UniGNN aggregation:

$$h_e = \phi_1(\{x_j\}_{j \in e}), \quad \tilde{x}_v = \phi_2(x_v, \{h_e\}_{e \in E_v}), \tag{17}$$

where ϕ_1 is mean aggregation and ϕ_2 is instantiated as UniGCN, UniGAT, or UniGIN. Final embeddings are layer-averaged:

$$\mathbf{g}_v = \frac{1}{K} \sum_{l=0}^{K-1} \mathbf{x}_v^{(l+1)} \quad \forall v \in \mathcal{V}. \tag{18}$$

We adopt pairwise ranking (BPR) on the hypergraph user-item scores:

$$L_{\text{global}} = -\sum_u \sum_{i \in H_u^+} \sum_{j \notin H_u^+} \ln \sigma(\mathbf{g}_u^\top \mathbf{g}_i - \mathbf{g}_u^\top \mathbf{g}_j). \tag{19}$$

For sequence $S_u = (i_1, \ldots, i_t)$:

$$\mathbf{x}_k = \mathbf{g}_{i_k} + \mathbf{p}_k, \quad \mathbf{h}_u = \text{Transformer}(\mathbf{x}_1, \ldots, \mathbf{x}_t)[t], \tag{20}$$

where $\mathbf{p}_k$ denotes the learnable positional embedding at position k. with loss

$$L_{\text{local}} = -\sum_{(u,t)} \log \frac{\exp(\mathbf{h}_u^\top \mathbf{g}_{i_{t+1}})}{\sum_j \exp(\mathbf{h}_u^\top \mathbf{g}_j)}. \tag{21}$$

Here, j indexes all candidate items in $\mathcal{I}$.

Standard Barlow Twins between the sequential state $\mathbf{h}_u$ and hypergraph user embedding $\mathbf{g}_u$:

$$\mathcal{L}_{\text{align}} = \sum_i (1 - C^{hg}_{ii})^2 + \lambda \sum_{i \neq j} (C^{hg}_{ij})^2. \tag{22}$$

Here, C^{hg} is the cross-correlation matrix between the batch-normalized representations $\tilde{\mathbf{h}}_u$ and $\tilde{\mathbf{g}}_u$.

Training follows the same two-phase schedule as CREATE-Pone.

At inference, both variants use only the sequential encoder with the enriched item embeddings ($\mathbf{z}_i$ for CREATE-Pone, $\mathbf{g}_i$ for CREATE-Uni). For a user representation $\mathbf{h}_u$, prediction scores are computed by inner product with enriched item embeddings, i.e., $\hat{y}_{u,i} = \mathbf{h}_u^\top \mathbf{e}_i$, where $\mathbf{e}_i = \mathbf{z}_i$ for CREATE-Pone and $\mathbf{e}_i = \mathbf{g}_i$ for CREATE-Uni.

IV. Theoretical Analysis

We organise the analysis into two parts. Section IV-A motivates CREATE-Pone by identifying two concrete deficiencies of unsigned LightGCN — *exposure bias* and *polarity blindness* — and explaining how the signed dual-branch construction and orthogonality alignment term each address them. Section IV-B justifies CREATE-Uni by invoking the established 1-WL / 1-GWL expressivity hierarchy [3], [15] and constructing an explicit user pair that LightGCN conflates but UniGNN separates.

A. CREATE-Pone: Signed Modeling Advantages

1) Setup and Notation: For every user u, the item universe $\mathcal{I}$ partitions into three disjoint subsets:

$$\mathcal{I} = \underbrace{\mathcal{O}_u}_{\text{observed positives}} \cup \underbrace{\mathcal{D}_u}_{\text{observed negatives}} \cup \underbrace{\mathcal{S}_u}_{\text{never observed}}, \tag{23}$$

where $\mathcal{O}_u$ denotes items the user has explicitly liked (e.g. ratings ≥ 4), $\mathcal{D}_u$ denotes items the user has explicitly disliked (e.g. ratings ≤ 3), and $\mathcal{S}_u$ denotes items the user has never interacted with at all. All three sets are disjoint by construction.

2) Limitation 1 — Exposure Bias in Unsigned BPR: Since LightGCN folds $\mathcal{D}_u$ into the positive graph, BPR samples negatives exclusively from $\mathcal{S}_u$ — items the user has simply never been exposed to. However, $\mathcal{S}_u$ contains no genuine dislike signal: these are items the user may well enjoy but has never encountered. In practice $|\mathcal{S}_u| \gg |\mathcal{D}_u|$ for most users, so the model is trained to rank observed items above a vast pool of potentially enjoyable unseen items, receiving noisy and misleading supervision throughout.

This is the *exposure bias*: the true negative evidence in $\mathcal{D}_u$ is never used as such, while $\mathcal{S}_u$ — which carries no genuine negative signal — is treated as the sole source of negative supervision.

Resolution in CREATE-Pone. The dual feedback loss $\mathcal{L}_{\mathrm{DF}}$ (Eq. 9) draws negatives exclusively from $\mathcal{D}_u \equiv E_n$ — items the user has explicitly disliked. Items from $\mathcal{S}_u$ are never used as negative evidence, ensuring that every negative training signal is semantically meaningful.

3) Limitation 2 — Polarity Blindness in LightGCN: Because LightGCN places all edges from $\mathcal{O}_u \cup \mathcal{D}_u$ into a single unsigned graph, it has no awareness that $\mathcal{D}_u$ represents negative feedback. Every item a user has rated — whether liked or disliked — is treated as a positive neighbour and contributes to the user embedding via message passing in exactly the same way.

This is the *polarity blindness* of unsigned graphs. The user embedding becomes an indiscriminate mixture of interest and disinterest signals, with no mechanism to separate them. As a result, items in $\mathcal{D}_u$ actively pull the user representation toward themselves during aggregation — the same direction as liked items. Any candidate item j similar to a disliked item $a \in \mathcal{D}_u$ will therefore receive an *inflated* recommendation score via the inner product $\hat{y}_{uj} = e_u^\top e_j$, since a's embedding has been folded into e_u as if it were a preference. The more disliked items a user has interacted with, the more corrupted their embedding becomes.

Resolution in CREATE-Pone. The dual-branch construction propagates interest embeddings z_u solely on G_p (built from $\mathcal{O}_u$) and disinterest embeddings v_u solely on G_n (built from $\mathcal{D}_u$). Since G_p and G_n share no edges, the two branches are completely isolated — items from $\mathcal{D}_u$ can never corrupt z_u, and the interest signal is preserved cleanly.

4) Remark — Pull–Push Duality in Alignment: The alignment loss (Eq. 15) plays two complementary roles. The Barlow Twins term $\|C^{hz} - I\|_F^2 \to 0$ acts as a **pull**, aligning the sequential state h_u toward the interest subspace of z_u, so the transformer inherits collaborative signals from $\mathcal{O}_u$. The orthogonality term $\mu\|C^{hv}\|_F^2 \to 0$ acts as a **push**, driving the cross-correlation between h_u and v_u to zero and thereby preventing disinterest signals from $\mathcal{D}_u$ from leaking into sequential recommendations.

No such push exists in the original unsigned CREATE, which operates on a single graph over $\mathcal{O}_u \cup \mathcal{D}_u$ and has no loss term to stop the sequential encoder from absorbing aversion signals. CREATE-Pone's alignment is therefore strictly richer.

B. CREATE-Uni: Hypergraph Expressivity

1) Established Results We Rely On: We use three results from the literature without reproving them:

- **GNNs $\leq$ 1-WL** [15]: Any message-passing GNN (including LightGCN) is at most as powerful as the 1-dimensional Weisfeiler-Leman (1-WL) graph isomorphism test. Two nodes that 1-WL cannot distinguish will receive identical embeddings from any such GNN.
- **UniGNN $\equiv$ 1-GWL** [3]: When the aggregation functions ϕ_1 (node-to-hyperedge) and ϕ_2 (hyperedge-to-node) are injective on multisets, UniGNN is equivalent in power to the 1-dimensional *Generalized* Weisfeiler-Leman (1-GWL) test on hypergraphs.
- **1-GWL $\geq$ 1-WL** [16]: 1-GWL is strictly more powerful than 1-WL on structures containing multi-way relationships, such as hyperedges of size ≥ 2, which pairwise graphs cannot natively represent.

What remains to be shown is that the specific user-modeling scenario in CREATE-Uni actually *instantiates* the strict gap — i.e., that there is a concrete, interpretable user pair that LightGCN conflates but UniGNN separates.

2) Proposition 1 — LightGCN Conflates Users with Identical Item Neighbourhoods: **Claim.** If two users u_1, u_2 have $N(u_1) = N(u_2)$ (the same set of interacted items) in the bipartite graph, LightGCN assigns them identical embeddings at every layer.

Proof. By induction on layer k.

Base case ($k = 0$): Users are initialised from the same distribution (standard practice); assume WLOG $e_{u_1}^{(0)} = e_{u_2}^{(0)}$.

Inductive step: Assume $e_{u_1}^{(k)} = e_{u_2}^{(k)}$ and every item $i \in N(u_1) = N(u_2)$ has the same embedding $e_i^{(k)}$ (same item node, same embedding). Then

$$\begin{aligned} e_{u_1}^{(k+1)} &= \sum_{i \in N(u_1)} \frac{e_i^{(k)}}{\sqrt{|N(u_1)||N(i)|}} \\ &= \sum_{i \in N(u_2)} \frac{e_i^{(k)}}{\sqrt{|N(u_2)||N(i)|}} \\ &= e_{u_2}^{(k+1)}. \end{aligned} \tag{24}$$

The layer-averaged final embeddings are therefore identical. Formally, this follows from $N(u_1) = N(u_2)$ and the fact that 1-WL cannot distinguish nodes with identical neighbourhood multisets [15].

3) Proposition 2 — UniGNN Separates Users that Share Item Neighbourhoods but Have Different Session Structure: **Claim.** There exist users u_1, u_2 with $N(u_1) = N(u_2)$ (so LightGCN conflates them) whose session structure differs, and UniGNN with injective aggregation assigns them distinct embeddings.

Construction. Let $\mathcal{I} = \{a, b, c, d\}$ with distinct initial embeddings.

- u_1 has **one session** $\{a, b, c, d\}$ — a single hyperedge $e_1 = \{a, b, c, d\}$.
- u_2 has **two sessions** $\{a, b\}$ and $\{c, d\}$ — hyperedges $e_2 = \{a, b\}$ and $e_3 = \{c, d\}$.

Both users have the same bipartite neighbourhood $N(u_1) = N(u_2) = \{a, b, c, d\}$.

LightGCN cannot separate them. By Proposition IV-B2, $e_{u_1}^{(k)} = e_{u_2}^{(k)}$ for all k.

UniGNN separates them. Using sum aggregation for ϕ_1:

$$h_{e_1} = e_a + e_b + e_c + e_d \quad \text{(one hyperedge, size 4)},$$
$$h_{e_2} = e_a + e_b, \quad h_{e_3} = e_c + e_d \quad \text{(two hyperedges, size 2 each)}.$$

Now ϕ_2 aggregates the *multiset of incident hyperedge embeddings*:

$$\tilde{e}_{u_1}^{(1)} = \phi_2\Big(e_{u_1}^{(0)},\ \underbrace{\{\{h_{e_1}\}\}}_{\text{1 element}}\Big),$$

$$\tilde{e}_{u_2}^{(1)} = \phi_2\Big(e_{u_2}^{(0)},\ \underbrace{\{\{h_{e_2}, h_{e_3}\}\}}_{\text{2 elements}}\Big).$$

The second arguments are multisets of *different cardinality* (1 vs. 2) and different elements. ϕ_2 maps distinct multisets to distinct outputs:

$$\tilde{e}_{u_1}^{(1)} \neq \tilde{e}_{u_2}^{(1)}.$$

UniGNN successfully distinguishes the two users.

Semantic interpretation. u_1 bought $\{a, b, c, d\}$ *together* (e.g., a complete skincare routine in one basket) — the co-purchase group is a meaningful collaborative signal. u_2 bought $\{a, b\}$ and $\{c, d\}$ on separate occasions — each session carries its own coherence signal. LightGCN collapses these into the same representation; UniGNN encodes the structural difference, yielding richer item embeddings g_i for the sequential encoder to align with.

4) Corollary — CREATE-Uni Provides Richer Alignment Targets than CREATE: **Statement.** The Barlow Twins alignment loss $\mathcal{L}_{\text{align}}$ (Eq. 22) conveys strictly more discriminative gradient signal to the sequential encoder under CREATE-Uni than under the original CREATE.

Proof sketch. By Proposition IV-B3, u_1 and u_2 receive different hypergraph embeddings: $g_{u_1} \neq g_{u_2}$. In a batch containing both users, the batch-normalised targets $\tilde{g}_{u_1} \neq \tilde{g}_{u_2}$ contribute different columns to the cross-correlation matrix C^{hg}, so the alignment loss generates informative gradients that push h_{u_1} and h_{u_2} apart.

Under LightGCN, $z_{u_1} = z_{u_2}$ (Proposition IV-B2), so both users produce identical alignment targets. The sequential encoder receives no gradient from $\mathcal{L}_{\text{align}}$ to differentiate h_{u_1} from h_{u_2}, even though their behavioral patterns differ. The mutual information transferred via alignment is therefore strictly greater in CREATE-Uni.

V. Datasets

We evaluate our method on five widely used recommendation datasets: **Amazon Beauty(amazon-reviews-2023)**, **Amazon Books(amazon-reviews-2023)**, **Yelp**, **Yambda-50M**, and **KuaiRand-1K**. All datasets are modeled as bipartite graphs $\mathcal{G} = (\mathcal{U}, \mathcal{I}, \mathcal{E})$, where $\mathcal{U}$ denotes users, $\mathcal{I}$ denotes items, and $\mathcal{E}$ denotes user–item interactions.

Amazon Reviews 2023 (Beauty, Books): In these datasets, user nodes correspond to Amazon customers and item nodes correspond to products (beauty products or books respectively). An edge (u, i) represents a user review or purchase interaction with an item. Edge features include the review rating and timestamp of the interaction. Item node features are derived from product metadata such as category, title, and textual description when available. User nodes typically do not contain explicit features and are represented via learned embeddings.

TABLE I: Dataset Statistics

Dataset	#Users	#Items	#Interactions
Amazon Beauty	729.6K	207.6K	6.6M
Amazon Books	776.4K	495.1K	9.5M
Yelp	41K	30K	2.1M
Yambda-50M	8.2K	181K	881K
KuaiRand-1K	1K	4M	11.7M

Yelp: In the Yelp dataset, user nodes represent Yelp users and item nodes represent local businesses. Edges correspond to user reviews of businesses. Edge attributes include rating scores and timestamps. Item node features are constructed from business metadata such as categories and attributes, while user nodes are represented implicitly through their interaction histories.

Yambda-50M: Yambda-50M is a large-scale implicit feedback dataset derived from Yahoo! music interactions. User nodes represent listeners and item nodes correspond to music tracks. Edges denote implicit listening events between users and tracks. Edge features primarily consist of timestamps indicating when the interaction occurred. No explicit user or item features are provided; representations are learned directly from the interaction graph.

KuaiRand-1K: KuaiRand-1K is a short-video recommendation dataset collected from the Kuaishou platform. User nodes represent viewers and item nodes correspond to short videos. Edges denote user exposure and interaction events such as clicks or watches. Edge features include timestamps and interaction signals (e.g., click or watch behavior). Item node features include video metadata such as category and content descriptors when available.

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